OPTIONAL CHALLENGE– BRANCH PREDICTION EXPERIMENT

1. Design Approach:

Designed a two-bit saturating counter-based branch predictor using Verilog HDL to predict whether a branch is taken or not taken. Implemented four prediction states: Strongly Not Taken, Weakly Not Taken, Weakly Taken and Strongly Taken. Added total branch and misprediction counters to monitor prediction performance.

2. Verification Steps:

- Created a separate testbench.
- Applied reset to initialize the branch predictor.
- Provided a sequence of taken and not-taken branch outcomes.
- Observed the predicted branch direction.
- Compared predicted outcomes with actual branch outcomes.
- Checked the total branch counter.
- Verified the misprediction counter.
- Simulated using ModelSim.

3. Results:

The testbench checks branch prediction using the sequence T, T, N, T, N, N, T. The predictor updates its state according to actual branch outcomes. Total branches and mispredictions are monitored, and the expected count is 7 branches and 5 mispredictions.

4. Issues Faced:

Understanding the two-bit saturating counter mechanism was challenging. Correctly updating prediction states based on actual branch outcomes is important. Accurate counting of branch mispredictions and synchronization with the clock were also required.

5. Conclusion:

Successfully designed and demonstrated a two-bit branch prediction module using Verilog HDL to monitor prediction accuracy and branch outcomes.